WORKED SOLUTION

STEP I 2000, Question 12

Probability and Statistics • Geometric Distribution

STEP I 2000, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q12

The key idea

In each case, ask: *given that every attempt so far has failed, what is the chance that the next one succeeds?* Then multiply the failure probabilities for the first $n-1$ attempts by the success probability on the nth. The three rules give three very different shapes of answer: one where the conditional chance grows, one where it stays fixed from the start, and one where it stays fixed only after the first go.

Throughout, let N be the number of attempts needed, so we want $\mathrm{P}(N = n)$ for $n = 1, 2, 3, \ldots$. Call the key that fits the door the *right* key.

Solution

Part (i)

Here I never repeat a key, so I am trying the keys in a uniformly random order. By symmetry, the right key is equally likely to be in any of the k positions of that order. Hence

$$\mathrm{P}(N = n) = \frac{1}{k} \quad \text{for } n = 1, 2, \ldots, k, \qquad \mathrm{P}(N = n) = 0 \text{ for } n > k.$$

If the symmetry argument feels too slick, multiply conditional probabilities instead. After j failures there are $k - j$ untried keys, one of them right, so for $1 \leq n \leq k$

$$\mathrm{P}(N = n) = \frac{k-1}{k} \cdot \frac{k-2}{k-1} \cdots \frac{k-n+1}{k-n+2} \cdot \frac{1}{k-n+1} = \frac{1}{k},$$

since the product telescopes.

Part (ii)

Now each attempt is a fresh, independent choice from all k keys, so each attempt succeeds with probability $\frac{1}{k}$, whatever has happened before. To need exactly n attempts,

the first $n-1$ must fail and the nth must succeed:

$$\mathrm{P}(N=n)=\left(\frac{k-1}{k}\right)^{n-1}\frac{1}{k}, \qquad n=1,2,3,\ldots$$

This is a geometric distribution with parameter $\frac{1}{k}$, and there is no upper limit on n.

Part (iii)

The first attempt succeeds with probability $\frac{1}{k}$, so $\mathrm{P}(N=1)=\frac{1}{k}$.

Suppose now that attempt m (for some $m \geq 1$) has failed. Then the key tried at attempt m was a wrong key. At attempt $m+1$ I choose uniformly from the other $k-1$ keys.

> **Ask yourself** Does the history before attempt m affect the chance of success at attempt $m+1$?

No. The only thing that matters is which key was excluded, and it is always a *wrong* key. So the $k-1$ keys on offer always include the right one, and attempt $m+1$ succeeds with probability $\frac{1}{k-1}$ and fails with probability $\frac{k-2}{k-1}$, whatever happened earlier.

So for $n \geq 2$: the first attempt fails (probability $\frac{k-1}{k}$), attempts $2,\ldots,n-1$ fail (each with probability $\frac{k-2}{k-1}$), and attempt n succeeds (probability $\frac{1}{k-1}$). Therefore

$$\mathrm{P}(N=n)=\frac{k-1}{k}\left(\frac{k-2}{k-1}\right)^{n-2}\frac{1}{k-1}=\frac{1}{k}\left(\frac{k-2}{k-1}\right)^{n-2}, \qquad n \geq 2,$$

together with $\mathrm{P}(N=1)=\frac{1}{k}$.

Interestingly, $\mathrm{P}(N=2)=\frac{1}{k}$ as well, the same as in part (i). (When $k=2$ the rule forces me to alternate keys, so N is 1 or 2, each with probability $\frac{1}{2}$; the formula gives this if we read 0^0 as 1.)

Checks

- Each distribution sums to 1. In (iii), $\frac{1}{k}+\frac{1}{k}\sum_{j\geq 0}\left(\frac{k-2}{k-1}\right)^j=\frac{1}{k}+\frac{1}{k}(k-1)=1$.
- The means behave sensibly: $\frac{k+1}{2}$ in (i), k in (ii), and $1+\frac{(k-1)^2}{k}$ in (iii). Not repeating the last key helps a little compared with (ii), but much less than never repeating any key.
- We checked all three formulas exactly for $k=2,\ldots,5$ and $n \leq 7$ by enumerating every sequence of choices with fractions. A simulation of 2×10^5 trials of (iii) with $k=5$ gave $\mathrm{P}(N=3)\approx 0.1496$, against the exact $\frac{1}{5}\cdot\frac{3}{4}=0.15$.

Common slips

- In (iii), using $\frac{k-2}{k-1}$ for the first attempt too. The first choice is from all k keys, so the first factor is $\frac{k-1}{k}$.
- In (i), forgetting to say that $\mathrm{P}(N = n) = 0$ when $n > k$.
- In (iii), thinking the excluded key could be the right one. It is only ever a key that has just failed.

Beyond STEP

Part (iii) is a small *Markov chain*. What happens next depends only on the current state (here, which key was tried last), not on the whole history. Spotting a state that makes the past irrelevant is the central trick in the theory of Markov chains, and it turns messy conditional probabilities into simple products. Grimmett and Welsh, *Probability: An Introduction*, introduces Markov chains at exactly this level.

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